QDA (Quadratic Discriminant Analysis)

LDA is a special case of QDA where $\Sigma_h \equiv \Sigma\ \forall h$. In the QDA case, we let the covariance matrices vary across groups, so that

$$f_h(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma_h|^{1/2}} \exp\left(-\tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_h)'\Sigma_h^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_h)\right).$$

And $\log \mathbb{P}(G = h \mid \boldsymbol{X} = \boldsymbol{x}) \propto \log f_h(\boldsymbol{x}) + \log \pi_h$

$$= -\frac{p}{2}\log(2\pi) - \tfrac{1}{2}\log(|\Sigma_h|) - \tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_h)'\Sigma_h^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_h) + \log \pi_h$$

$$\propto -\tfrac{1}{2}\log(|\Sigma_h|) - \tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_h)'\Sigma_h^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_h) + \log \pi_h \equiv \delta_h(\boldsymbol{x}),$$

where $\delta_h(\boldsymbol{x})$ is called the *quadratic discriminant function*.

The decision boundary between any class h and ℓ is such that

$$\log\left(\frac{\mathbb{P}(G = h \mid \boldsymbol{X} = \boldsymbol{x})}{\mathbb{P}(G = \ell \mid \boldsymbol{X} = \boldsymbol{x})}\right) = 0, \qquad \text{i.e.} \qquad \{\boldsymbol{x} : \delta_h(\boldsymbol{x}) = \delta_\ell(\boldsymbol{x})\}.$$

The estimators for QDA are similar to LDA except for the covariance matrices. In the LDA case, we had

$$\hat{\Sigma}_h \equiv \hat{\Sigma} = \sum_{h=1}^{K}\sum_{i\in h}(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_h)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_h)'/(N-K), \quad \forall h,$$

in the QDA we now have

$$\hat{\Sigma}_h = \sum_{i\in h}(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_h)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_h)'/(N_h - 1).$$

When p is large this could lead to an important increase in parameters. With LDA we had

$$\underbrace{Kp}_{\text{class means}} + \underbrace{\frac{p(p+1)}{2}}_{\substack{\text{common covariance} \\ \text{matrix (symmetric)}}} + \underbrace{K-1}_{\substack{\text{class probabilities} \\ \text{(as they sum to one} \\ \text{we only need } K-1 \text{ of them)}}},$$

while for QDA it is

$$\underbrace{Kp}_{\text{class means}} + \underbrace{K\frac{p(p+1)}{2}}_{K \text{ covariance matrices}} + \underbrace{K-1}_{\text{class probabilities}}.$$